AlN Measured Absorption: Before / After

`shg_mode_survey · loaded_sweep1 + trial_grid + wide_window`

413 refined crossings across 26 geometries, reprocessed with AlN's own measured bulk absorption (AM187-LT6 ellipsometry, extraordinary-axis k_e, added to the existing loss budget -- see geometry.aln_core_loss_dB_per_m). AlN was assumed lossless in every result before 2026-10-12.

NCE (eta_pct_per_W_cm2) does not depend on loss at all -- it is EXACTLY unchanged. Only loss-limited PEAK efficiency (which optimizes device length against both pump and SH loss) can move, since adding AlN absorption both raises the loss rate and shortens the optimal device length.

PEAK EFFICIENCY DROP FACTOR -- RIDGE FAMILY
1.9e5x median, 1.5e7x max
bare AlN ridge had ~zero SH-side loss before -- hit hardest

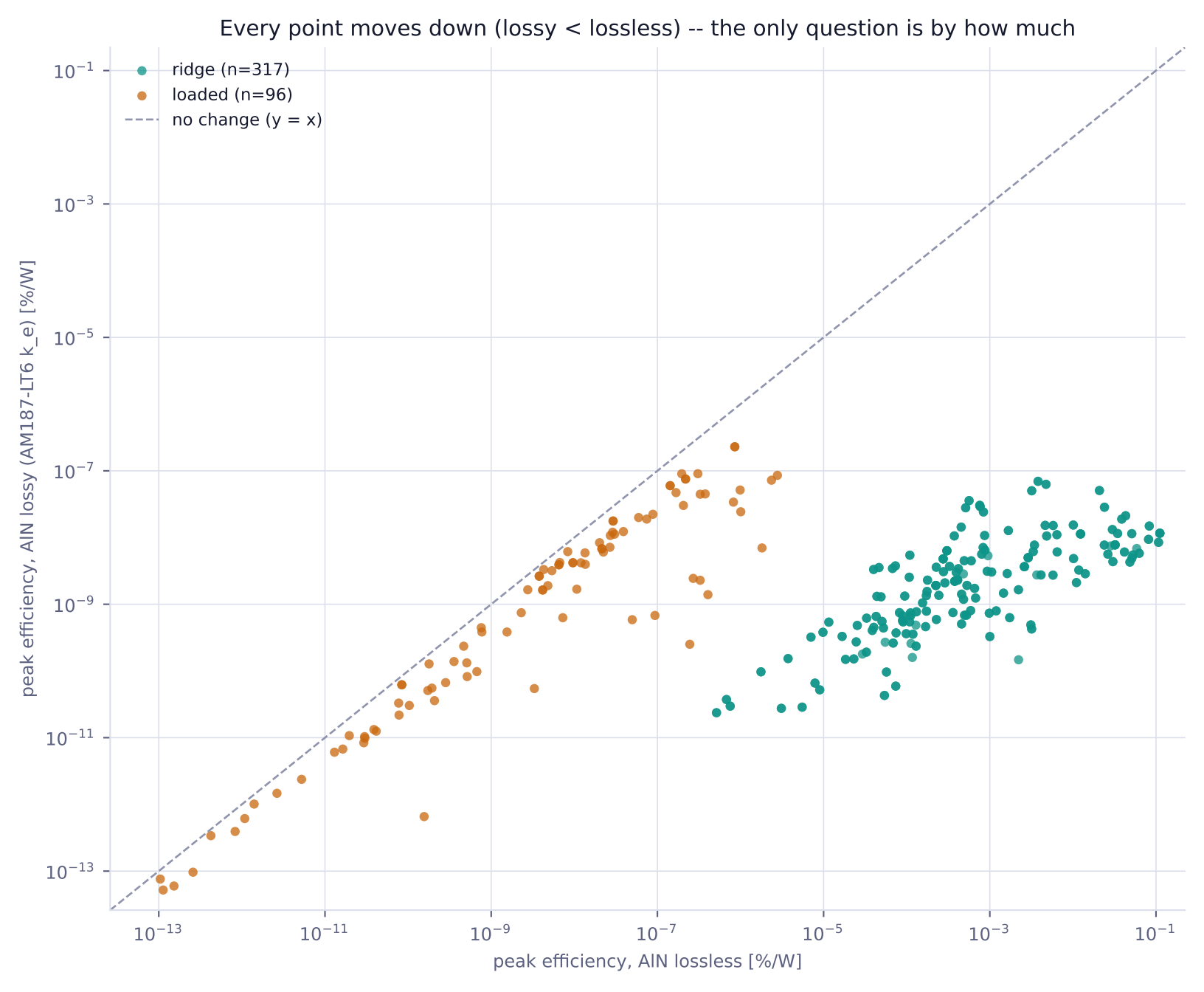
PEAK EFFICIENCY DROP FACTOR -- LOADED FAMILY
3x median, 976x max
already had a lossy TiO2 strip -- AlN adds a smaller relative penalty

ALN SH-BAND LOSS VS. EXISTING TOTAL
28205% added
median, relative to the pre-existing SH loss budget

ALN PUMP-BAND LOSS VS. EXISTING TOTAL
44207% added
median, relative to the pre-existing pump loss budget

CAUTION

k_e (extraordinary axis) only, applied uniformly wherever the mode has power in the AlN core -- not field-direction-resolved against k_o (2026-10-12 decision: the existing confinement-weighting machinery is polarization-agnostic). pedestal family excluded -- no field exports available, see this file's module docstring.


Every point moves down (lossy < lossless) -- the only question is by how much
ridge (n=317)
loaded (n=96)
no change (y = x)
peak efficiency, AlN lossy (AM187-LT6 k_e) [%/W]
peak efficiency, AlN lossless [%/W]
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}

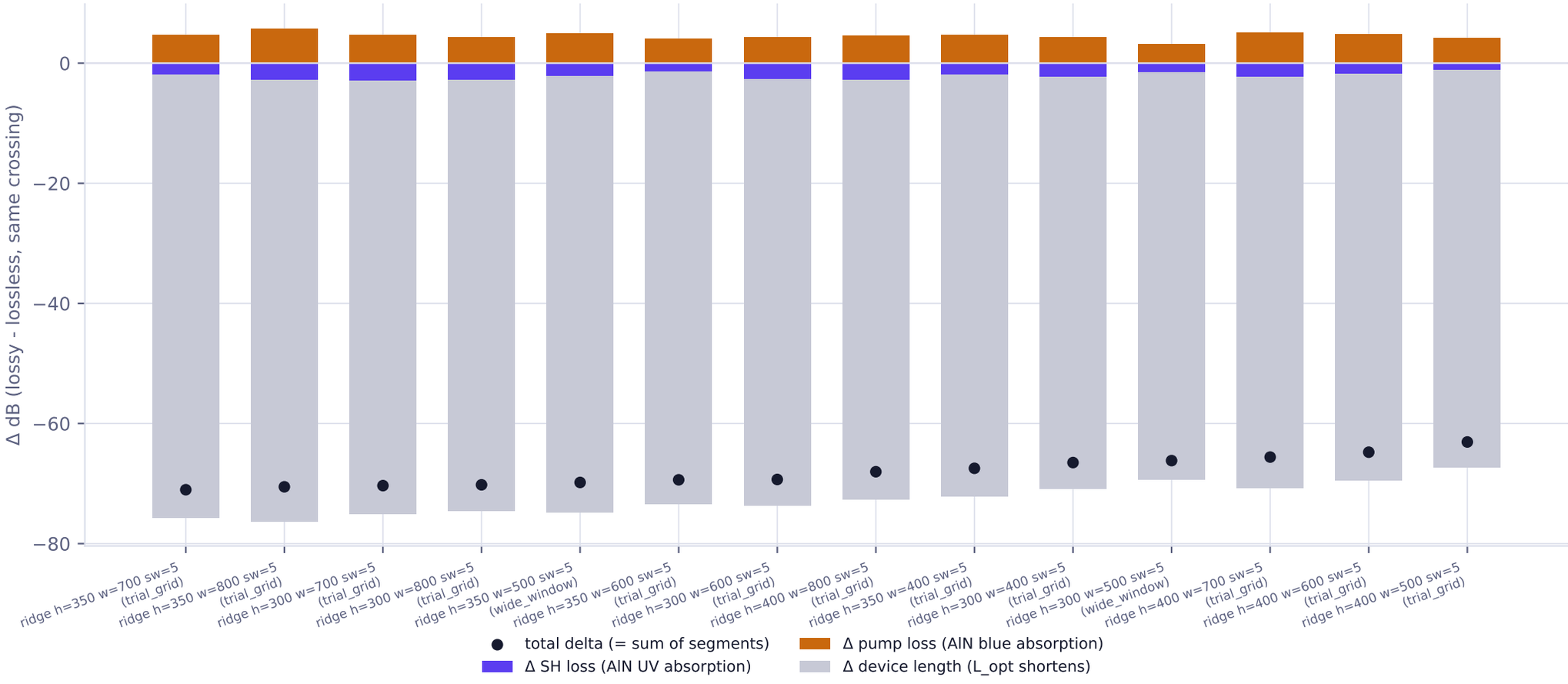
Top 15 crossings by lossless peak efficiency -- where do they rank once AlN absorbs?

run · geometry · pair

#	GEOMETRY	PUMP→SH	LOSSLESS [%/W]	LOSSY [%/W]	NEW RANK
1	ridge h=350 w=500 sw=5 (wide_window)	TM00→TM04	0.111	1.16e-08	**#66 ↓**
2	ridge h=350 w=500 sw=5 (wide_window)	TM00→TM04	0.111	1.16e-08	**#65 ↓**
3	ridge h=350 w=500 sw=5 (trial_grid)	TM00→TM04	0.111	1.16e-08	**#64 ↓**
4	ridge h=350 w=500 sw=5 (trial_grid)	TM00→TM04	0.111	1.16e-08	**#63 ↓**
5	ridge h=350 w=700 sw=5 (trial_grid)	TM00→TM04	0.107	8.47e-09	**#87 ↓**
6	ridge h=350 w=700 sw=5 (trial_grid)	TM00→TM04	0.107	8.47e-09	**#88 ↓**
7	ridge h=350 w=400 sw=5 (trial_grid)	TM00→TM04	0.083	1.49e-08	**#53 ↓**
8	ridge h=350 w=400 sw=5 (trial_grid)	TM00→TM04	0.083	1.49e-08	**#54 ↓**
9	ridge h=350 w=600 sw=5 (trial_grid)	TM00→TM04	0.0813	9.39e-09	**#85 ↓**
10	ridge h=350 w=600 sw=5 (trial_grid)	TM00→TM04	0.0813	9.39e-09	**#86 ↓**
11	ridge h=300 w=700 sw=5 (trial_grid)	TM00→TM04	0.0628	5.8e-09	**#122 ↓**
12	ridge h=300 w=700 sw=5 (trial_grid)	TM00→TM04	0.0628	5.8e-09	**#121 ↓**
13	ridge h=300 w=600 sw=5 (trial_grid)	TM00→TM04	0.0586	6.85e-09	**#103 ↓**
14	ridge h=350 w=700 sw=5 (trial_grid)	TM10→TM04	0.0529	5.48e-09	**#127 ↓**
15	ridge h=350 w=700 sw=5 (trial_grid)	TM10→TM04	0.0529	5.48e-09	**#128 ↓**

15 of the original top 15 fall out of the top 15 once AlN's own absorption is included (orange, marked with new rank).

Why peak efficiency drops: exact dB delta per geometry (worst first)



Each bar is the SAME crossing's best-by-lossless-peak-efficiency result, decomposed twice (nce_decomposition.decompose, include_aln=False/True) -- only sh_loss_dB, pump_loss_dB and length_dB can differ (shape_dB/field_dB depend only on overlap_shape and eta_pct_per_W_cm2, neither of which depends on loss). The black dot is the exact sum of the three segments, verified against the direct peak-efficiency ratio.

Reality check: what SH loss is consistent with measured device output?

`ridge h=350 w=500 nm, TM00→TM04 -- closest match to the measured device`

Given a real device's actual, fixed propagation length (not an optimized L_opt) and actual pump power reaching the waveguide, shg_physics.loss_limited_length's own model (P_SH = eta_SI * P_pump^2 * L_eff(L; alpha_p, alpha_s)^2) can be inverted for the SH-band loss that reproduces a measured SH output -- independent of whether any particular ellipsometry sample's k is trustworthy. Using this geometry's own simulated eta_SI and currently-modeled native pump loss (15.8 dB/cm, no AlN):

PUMP POWER (IN WAVEGUIDE)	INTERACTION LENGTH	TARGET SH OUTPUT	IMPLIED alpha_SH BOUND
100 mW	2000 um (2 mm)	20 nW	**<= 679 dB/cm**
100 mW	2000 um (2 mm)	200 nW	**<= 235 dB/cm**

AM187-LT6 k_e implies 100,161 dB/cm for this same crossing -- 147x to 427x above what the device's own measured output permits.

20 nW is the user's suggested representative floor (most devices clear at least this much; 200 nW was closer to the best-observed case, shown for context). Pump power is fixed at the 100 mW (50% of the 200 mW incident, the optimistic end of the 10-50% coupling-efficiency uncertainty) case specifically requested for this reality check -- see the earlier scan (10/25/50% coupling) for the full sensitivity. This is a bound, not a measurement of alpha_SH itself: it assumes the modeled eta_SI and native pump loss for this geometry are correct, and that the real device isn't compensating for higher SH loss some other way (e.g. a shorter effective interaction length than the nominal 2 mm, imperfect mode matching reducing the EFFECTIVE eta).

Notes & provenance

Data: runs/loaded_sweep1/crossings.csv, runs/trial_grid/crossings.csv, runs/wide_window/crossings.csv, refined rows with a reprocessed AlN-loss pair only (aln_pump_loss_dB_per_m not null). Generated via build_aln_loss_report.py / nce_decomposition.py.

AlN's own bulk absorption: geometry.aln_core_loss_dB_per_m(wavelength_nm), a Beer-Lambert conversion of the AM187-LT6-measured k_e (extraordinary axis, 3-Lorentzian fit -- see materials/fit_aln_measured_k.py), confinement-weighted by the mode's power fraction in geom.core_mask() -- same first-order-perturbation pattern already used for shell/strip bulk loss (survey.py's bulk_absorption_loss). k_e only, not field-direction-resolved against k_o (k_e is ~90x k_o and is the axis d33 couples -- Py = d33*Ey^2 -- so this is the conservative/dominant choice, not a field-weighted split; see PLAN.md 2026-10-12).

Computed entirely from each refined crossing's already-exported field data (reprocess_run.py) -- no EMode re-solve was needed for any row shown here. Original AlN-lossless columns (pump_absorption_dB_per_m / sh_absorption_dB_per_m / peak_efficiency_pct_per_W) are untouched in crossings.csv; AlN-lossy figures here are computed downstream, on the fly.

Excluded: pedestal1/pedestal_sweep1/pedestal_vertical1 (zero field exports on either machine; re-solving from scratch was estimated at ~9.5 hours of EMode time for a geometry already being superseded by the corrected embedded 'hybrid' family -- not worth it, per 2026-10-12 decision). trial_grid is missing 11/280 refined crossings' exports even after merging the laptop's local runs folder -- those rows are excluded here too, not zero-filled.